FIE401 - Assignment 2
Estimating market risk premia

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Overview

A central question in asset pricing is how to value an asset. Put differently, how to quantify the risk-return trade-off. Development of the CAPM allowed economists to formally quantify the risk (market risk = β) and the reward for carrying it (market risk premia = λ).

In this assignment, you are going to estimate market risk (i.e., β) of 25 portfolios sorted on size and book-to-market ratio based on the CAPM model. Afterwards, you are going to estimate market risk premia implied by the CAPM model. This approach is known as Fama-MacBeth (1973) approach.

Formalities

This assignment will be handed out on 10.03.2026 at 12:00 and has to be submitted no later than the **17.03.2026 at 09:00**. Submit your commented coding file (file extension: ".R") and a pdf file (file extension: ".pdf") including the numerical results and answers to questions posted.

Please comment your code shortly so that a reader can reconstruct your thinking. You do not need to explain the used functions. You do not need to describe your coding in the pdf file. Please keep your answers very brief. **Code without comments will automatically result in a "Fail" grade.**

On Canvas, you will find a guideline of how to make good tables. Note that in the last lecture of this course, we will cover the subject of how to present research in detail.

After submission, the assignments will be randomly redistributed. Read your peer's work carefully and comment within Canvas no later than **24.03.2026 at 09:00**. Your comments are meant to improve the work of your peers, so use constructive feedback. See peer-review guidelines for details. **Insufficient quality of peer-review or no peer-review will result in a "Fail" grade, regardless of the quality of your own analysis.**

In order to make it easy for us to allocate individual assignment submission as well as peer review to groups, please follow following file name structure. **For the assignment submission: "Assignment2__group__XX.R/pdf".** "XX" indicates the group number that submitted the assignment. **Please submit the peer review as a comment on CANVAS.**

Please do not include any personal information such as name, student number, etc. in the submitted files.

Note that in order to complete the assignment, you need to submit your own work as well as complete the peer review. Both aspects will be reviewed.

Work together in groups of 3-4 people.

Submit your assignment even if you do not finish all tasks.

Note that some concepts are introduced for the first time in this assignment. Please use online resources actively.

We touched on almost all functions you have to use to solve this assignment. If you are unsure how to use a function, either use R's own documentation (type ?command) or use www.stackoverflow.com. We encourage you to code this assignment yourselves, and not to use purpose-made solutions or packages as provided on the internet.

Before starting to code, try to separate the task into smaller pieces.

Data description

File 25_portfolios_on_size_BM.Rdata contains monthly returns on 25 portfolios sorted on size (ME) and book-to-market ratio (BM) as well as market excess return (MktRF) and risk free rate (RF) from January 1980 till December 2017. ME1 contains the smallest stocks, while ME5 contains the largest stocks. BM1 contains stocks with the lowest book-to-market ratio, while BM5 contains stocks with the highest book-to-market ratio. Data come from Kenneth French's website.

Tasks

First look at the data

- Load the data and investigate it.

Fama-MacBeth approach: Part 1

- For each portfolio p estimate market risk (β_p) implied by CAPM:

$$R_{pt} - RF_t = \alpha_p + \beta_p \times MktRF_t + u_{pt}$$

- This step should result in having 25 market betas ($\widehat{\beta_p}$) for each of the 25 portfolios from 25 individual regressions. Note, that I provide a coding hint on how to use loops and save regression coefficients at the end of the assignment.
- Plot the market beta of each individual portfolio on the x-axis against the average return over the period on the y-axis. Does market risk explain differences in return? Does the sign of the relationship (positive/negative) correspond to your expectations?

Fama-MacBeth approach: Part 2

- Estimate market risk premia (λ).
- For each month run cross-sectional regression of portfolio returns on $\widehat{\beta_p}$:

$$R_{pt} = a_t + \lambda_t \times \widehat{\beta_p} + \omega_{pt}$$

- This step should result in having 456 risk premias ($\widehat{\lambda_t}$) for each of the 456 months in the sample from 456 individual regressions. Note, that I provide a coding hint on how to use loops and save regression coefficients at the end of the assignment.
- Plot time-series of monthly $\widehat{\lambda_t}$.

Fama-MacBeth approach: Part 3

- Compute average of monthly $\widehat{\lambda_t}$ and test if it is significantly different from 0.
- What do you conclude regarding the price of market risk?
- Which bias might you be concerned about in this analysis, and how would it affect your results?

Coding hint

In two tasks of this assignment, you are asked to run sequential regressions and save the coefficients. Here I provide a general **pseudo-code** of how this can be done.

```
# object to save 25 coefficients
coef.vec <- rep(NA, 25)

# iteration
for(i in ...){
  # construct relevant data
  rel.data <- ...

  # regression
  fit <- lm(...., data = rel.data)

  # save coefficient
  coef.vec[i] <- fit$coef[...]
}
```

Note, feel free to use more elegant and faster coding approaches.

References

Fama, E. F., MacBeth, J. D., 1973, Risk, Return and Equilibrium: Empirical Tests. Journal of Political Economy 81 (3), pp. 607-636